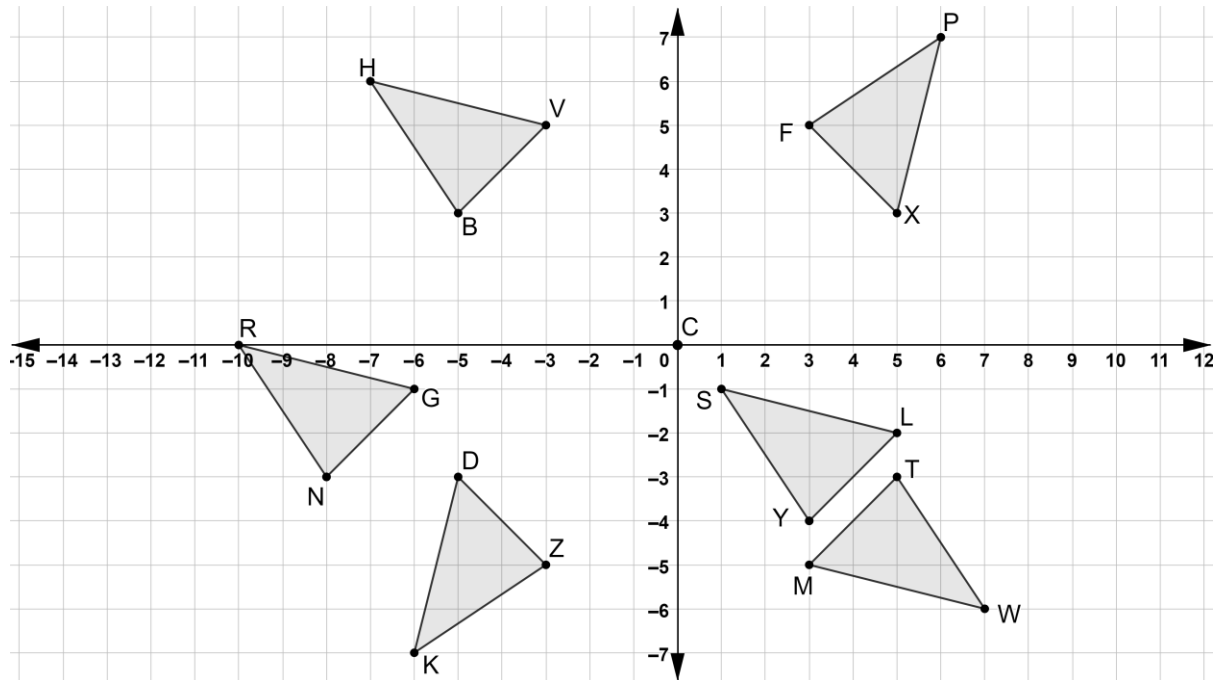


Six triangles are shown on the coordinate grid.



1. Draw the mapping that shows the translation if $\triangle SYL$ is the preimage and $\triangle HBV$ is the image.
2. Draw the mapping that shows the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.
3. Describe the translation if $\triangle HBV$ is the preimage and $\triangle RNG$ is the image.

4. Complete the statements.

The x -values of $\triangle RNG$, the image, can be found by

- ☐ adding
☐ subtracting

___ unit(s) to the x -values of $\triangle SYL$, the preimage. The y -values

of $\triangle RNG$, the image, can be found by

- ☐ adding
☐ subtracting

___ unit(s)

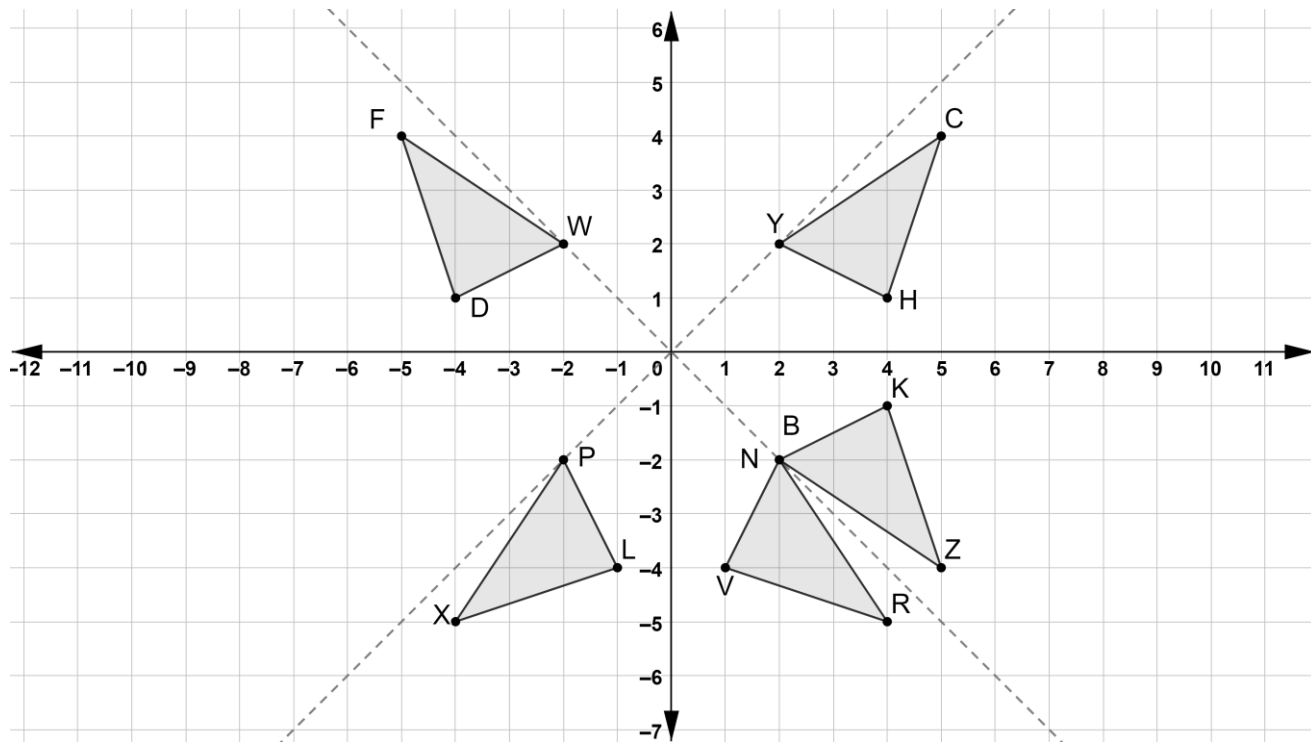
to the y -values of $\triangle SYL$, the preimage.

5. On the coordinate grid draw angle DCV .
6. The algebraic description of the rotation of $\triangle KZD$, the preimage, onto $\triangle HBV$ is the image, is $(x, y) \rightarrow (y, -x)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.
7. The algebraic description of the rotation of $\triangle TWM$, the preimage, onto $\triangle BHV$ is the image, is $(x, y) \rightarrow (-x, -y)$. Describe the rotation. Then, use the coordinates of the vertices to show how the rotation matches the algebraic description.
8. Describe how the measure of $\angle BCF$ can be used to determine the degrees of rotation where $\triangle PFX$ is the preimage and $\triangle HBV$ is the image.
9. Describe how the coordinates of the vertices of $\triangle PFX$, the preimage, and the coordinates of $\triangle HBV$, the image can be used to determine the degrees of rotation.
10. Describe how the orientation of the triangles helps to determine which transformations were a translation and which transformations were a rotation.

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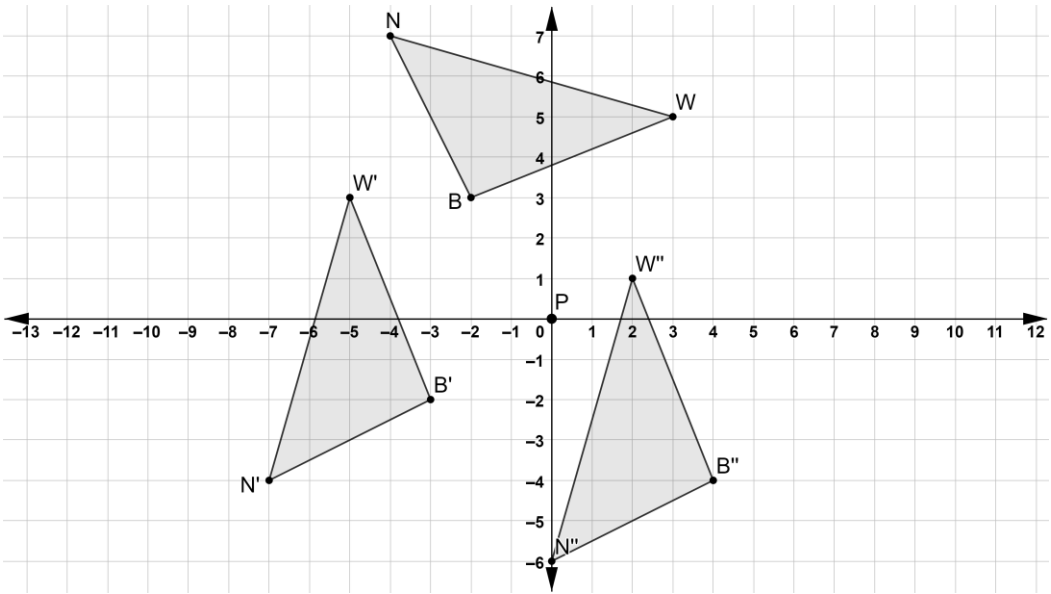
Six triangles are shown on the coordinate grid.



- Draw the reflection of $\triangle PLX$ across the line $y = x$. Label the reflection as $\triangle P'L'X'$.
- Draw the reflection of $\triangle FWD$ across the line $y = -x$. Label the reflection as $\triangle F'W'D'$.
- Select all of the triangles that are reflections of $\triangle CHY$.
 - ☐ $\triangle FDW$
 - ☐ $\triangle XLP$
 - ☐ $\triangle RVN$
 - ☐ $\triangle ZKB$
 - ☐ $\triangle P'L'X'$
 - ☐ $\triangle F'W'D'$

4. The algebraic description of the reflection of $\triangle BKZ$, the preimage, onto $\triangle NVR$, the image, is $(x, y) \rightarrow (-y, -x)$. Describe how the coordinates of the vertices of $\triangle BKZ$, the preimage, and the coordinates of $\triangle NVR$ the image can be used to determine the line of reflection.
5. Which triangle is the image of reflecting $\triangle FDW$ across the x -axis?
6. Which triangle is the image of reflecting $\triangle CHY$ across the y -axis?
7. Describe how the coordinates of the vertices of $\triangle F'W'D'$, the preimage, and the coordinates of $\triangle XPL$ the image can be used to determine the algebraic description of the reflection.
8. Describe how the coordinates of the vertices of $\triangle ZKB$, the preimage, and the coordinates of $\triangle CHY$, the image can be used to determine the algebraic description of the reflection.
9. Write the algebraic description for the reflection of $\triangle NVR$, the preimage, onto $\triangle PLX$, the image.
10. Write the algebraic description for the reflection of $\triangle P'L'X'$, the preimage, onto, $\triangle BKZ$, the image.

Triangles BWN , $B'W'N'$, and $B''W''N''$ are shown on the coordinate grid where $\triangle BWN$ is the preimage.



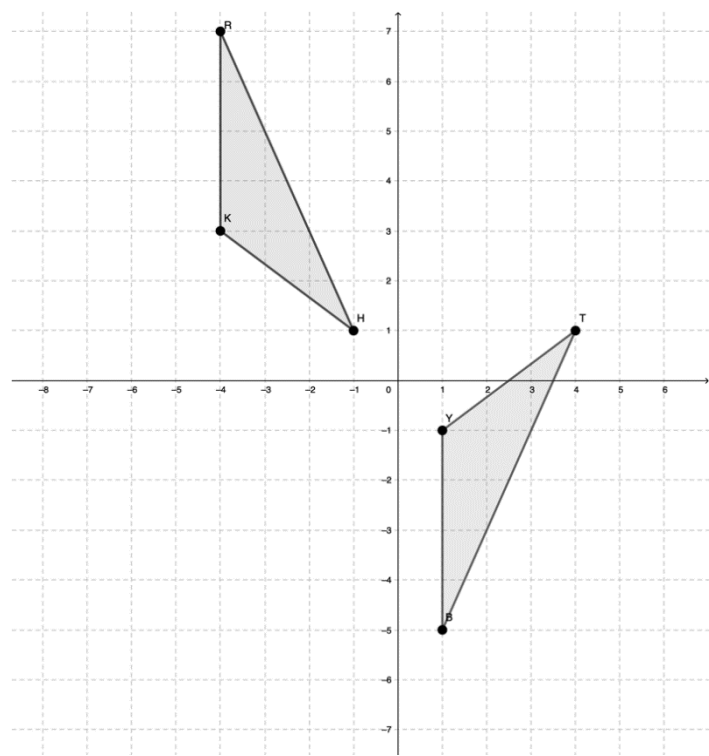
- Describe the feature of $\triangle B'W'N'$ that indicates the rigid motion that was applied to $\triangle BWN$ that resulted in $\triangle B'W'N'$.
- Show the feature described by mapping the transformation of one vertex of $\triangle BWN$ to $\triangle B'W'N'$.
- Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle BWN \rightarrow \triangle B'W'N'$	

- Map the transformation of one vertex of $\triangle B'W'N'$ to $\triangle B''W''N''$.
- Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle B'W'N' \rightarrow \triangle B''W''N''$	

Triangles RKH and BYT are shown on the coordinate grid where $\triangle RKH$ is the preimage.

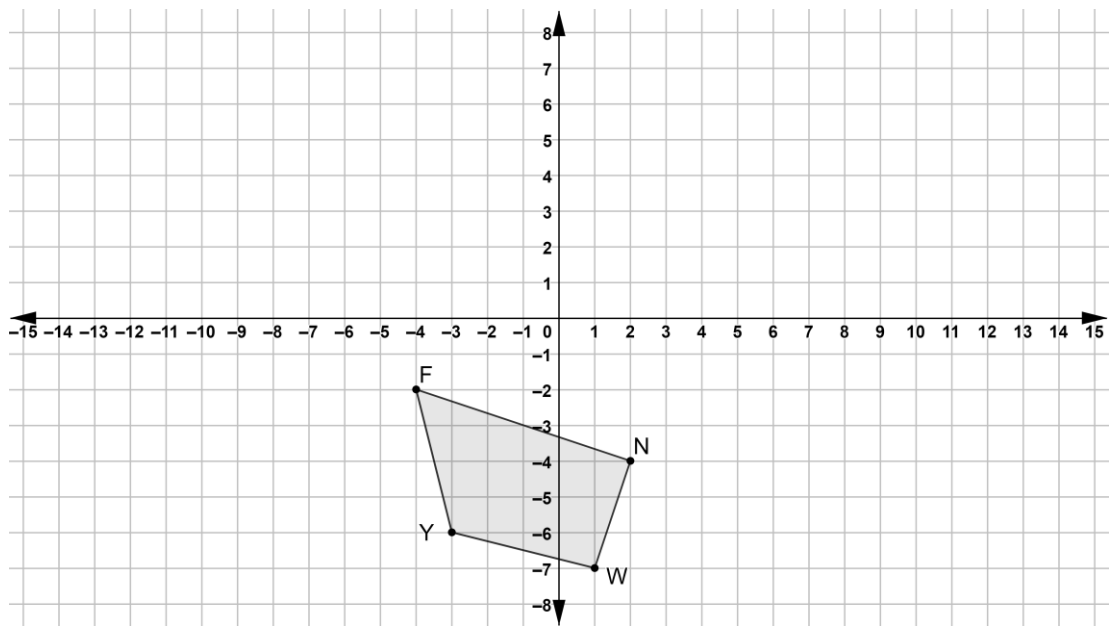


- 6. Describe the feature of $\triangle TYB$ that indicates that more than one rigid motion was applied to $\triangle RKH$.
- 7. Draw $\triangle R'K'H'$, the result of the first transformation that was applied to $\triangle RKH$.
- 8. Map the transformation of one vertex of $\triangle R'K'H'$ to $\triangle BYT$.
- 9. Complete the table.

Transformation	Mapping	Algebraic Description
	$\triangle RKH \rightarrow \triangle R'K'H'$	
	$\triangle R'K'H' \rightarrow \triangle BYT$	

- 10. Explain how to know if the sequence of transformations preserves distance.

Quadrilateral $FNWY$ is shown on the coordinate grid.



1. Draw the quadrilateral that is the result of a translation left 4 units and up 8 units and then rotated 90° clockwise about the origin. Label the new quadrilateral $GLTX$.

Complete the table.

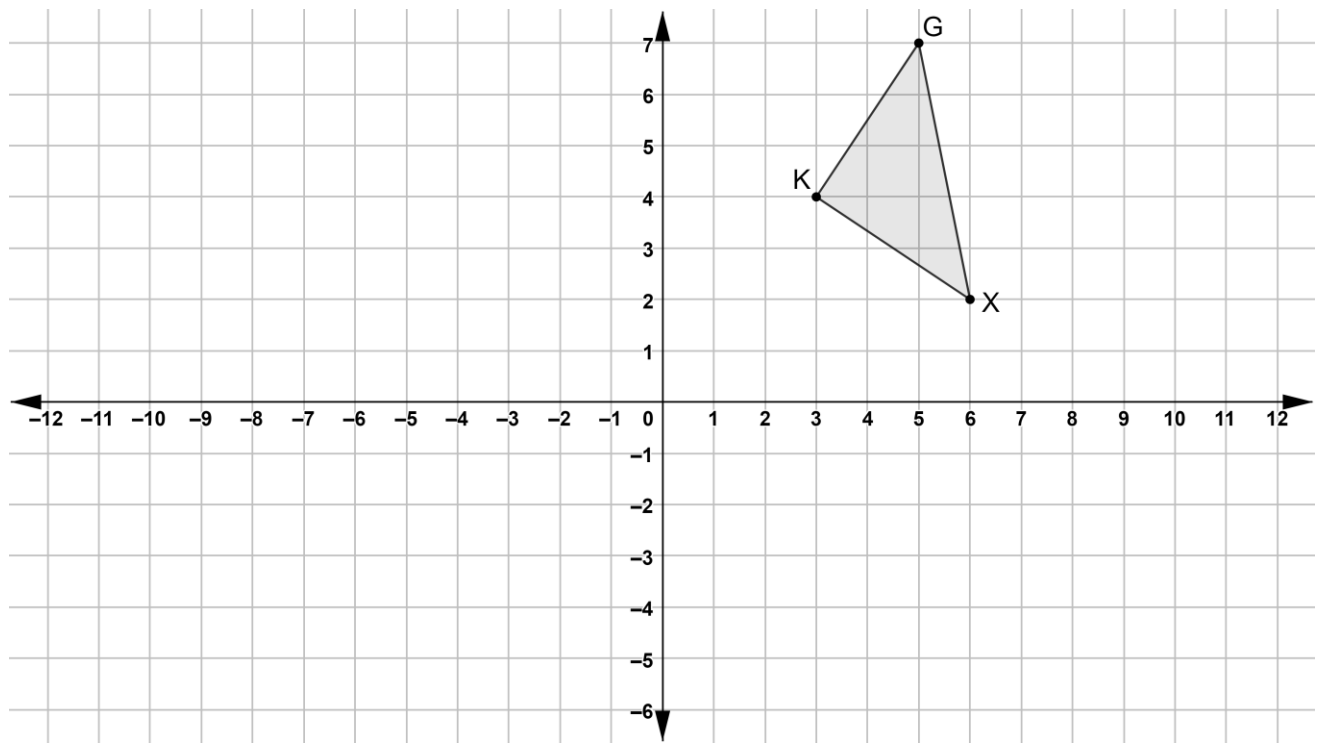
Transformation	Mapping	Algebraic Description
2.	$FNWY \rightarrow F'N'W'Y'$	
3.	$F'N'W'Y' \rightarrow GLTX$	

4. Complete the table of values for each quadrilateral.

Quadrilateral $FNWY$			Quadrilateral $F'N'W'Y'$			Quadrilateral $GLTX$		
Vertex	x	y	Vertex	x	y	Vertex	x	y
F			F'			G		
N			N'			L		
W			W'			T		
Y			Y'			X		

- Explain how the tables for quadrilaterals $FNWY$ and $F'N'W'Y'$ show the effect of the transformation on the coordinates that result in the algebraic description.
- Explain how the tables for quadrilaterals $F'N'W'Y'$ and $GLTX$ show the effect of the transformation on the coordinates that result in the algebraic description.

Triangle XGK is shown on the coordinate grid.



- Draw the triangle that is the result of a rotation 270° counterclockwise about the origin and then a reflection across the y -axis. Label the new triangle $X''G''K''$.

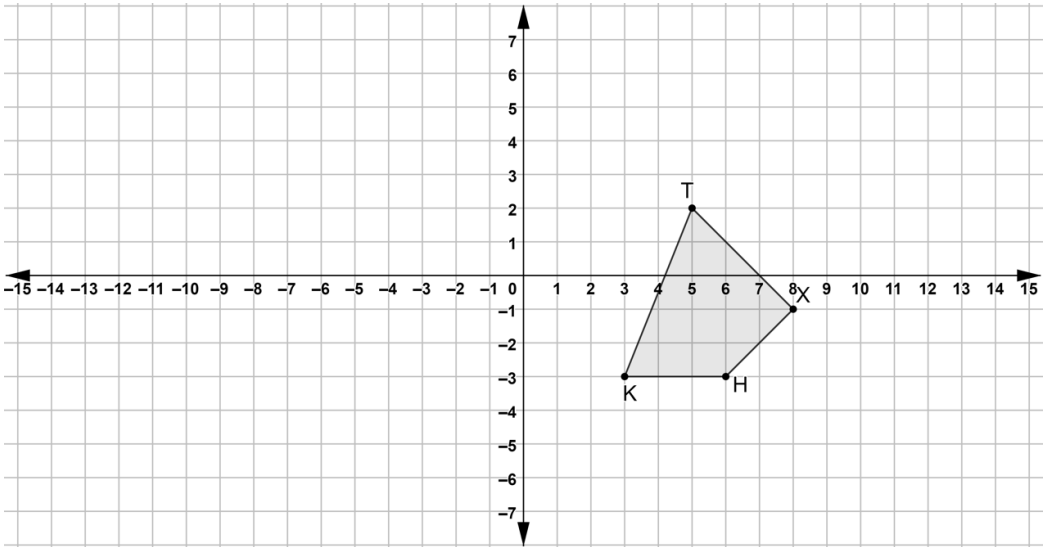
Complete the table.

	Transformation	Mapping	Algebraic Description
8.		$\triangle XGK \rightarrow \triangle X'G'K'$	
9.		$\triangle X'G'K' \rightarrow \triangle X''G''K''$	

- Describe the connection between the algebraic description of the rotation and the vertices of $\triangle XGK$ and $\triangle X'G'K'$.

Quadrilateral $TXHK$ is shown on the coordinate grid.

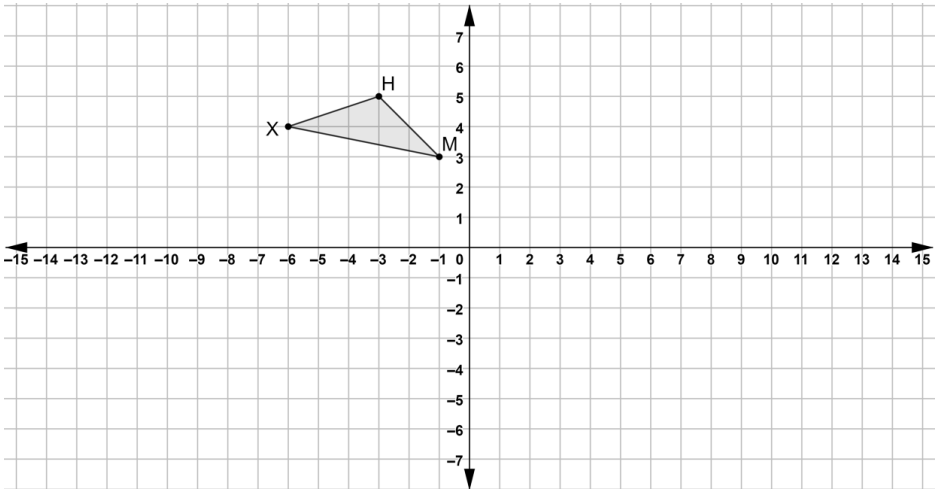
Perform the series of transformations on quadrilateral $TXHK$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
1.	$(x, y) \rightarrow (y, -x)$	$T'X'H'K'$
2.	$(x, y) \rightarrow (x - 3, y + 5)$	$T''X''H''K''$

Triangle XHM is shown on the coordinate grid.

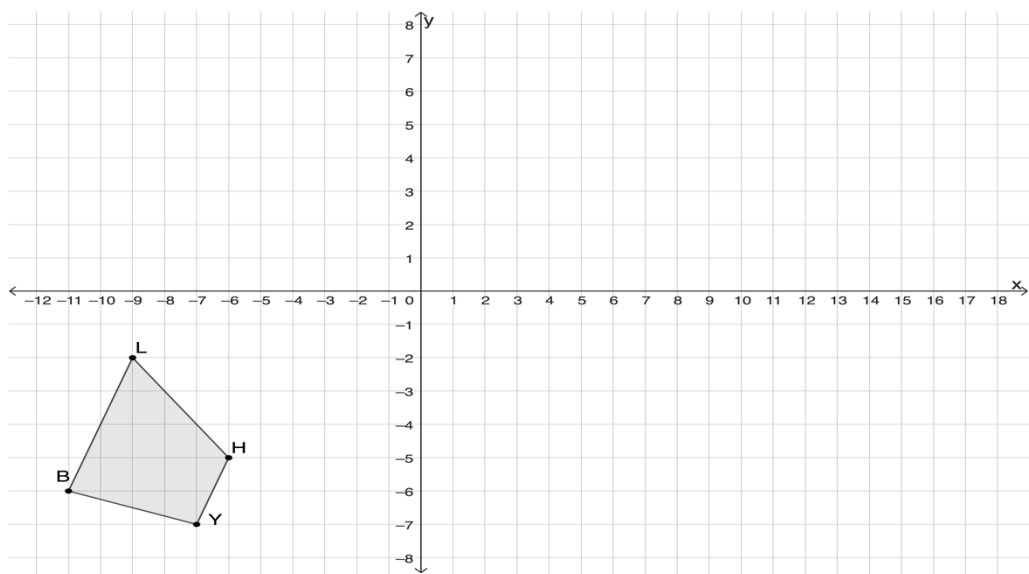
Perform the series of transformations on $\triangle XHM$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Triangle
3.	$(x, y) \rightarrow (x + 10, y - 8)$	$X'H'M'$
4.	$(x, y) \rightarrow (x, -y)$	$X''H''M''$

Quadrilateral $LHYB$ is shown on the coordinate grid.

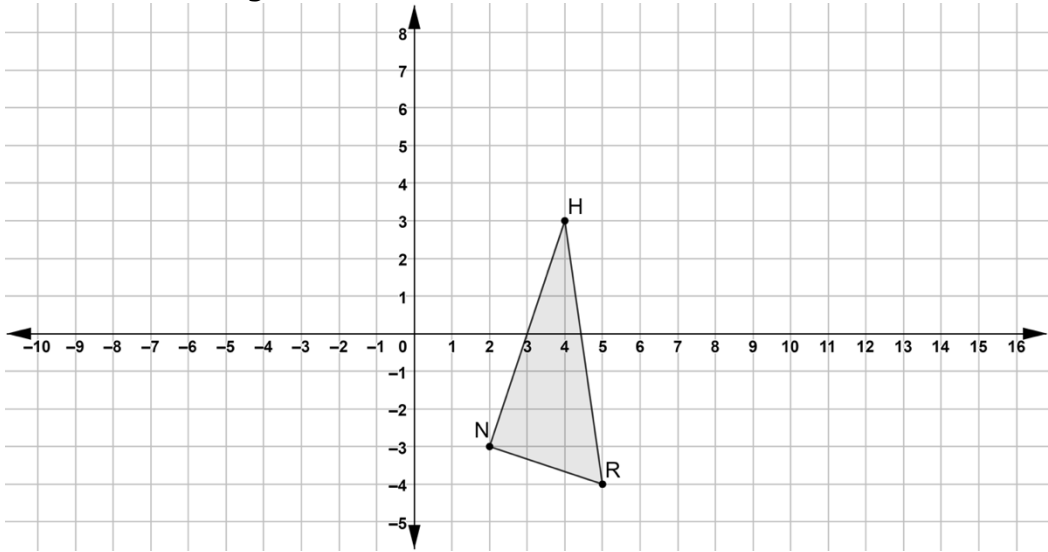
Perform the series of transformations on quadrilateral $LHYB$. Assume all rotations are centered at the origin. Label each result.



	Transformation	Quadrilateral
5.	$(x, y) \rightarrow (-x, -y)$	$L'H'Y'B'$
6.	$(x, y) \rightarrow (x, -y)$	$L''H''Y''B''$
7.	$(x, y) \rightarrow (x + 5, y + 7)$	$L'''H'''Y'''B'''$

Triangle NHR is shown on the coordinate grid.

Perform the series of transformations on $\triangle NHR$. Assume all rotations are centered at the origin. Label each result.

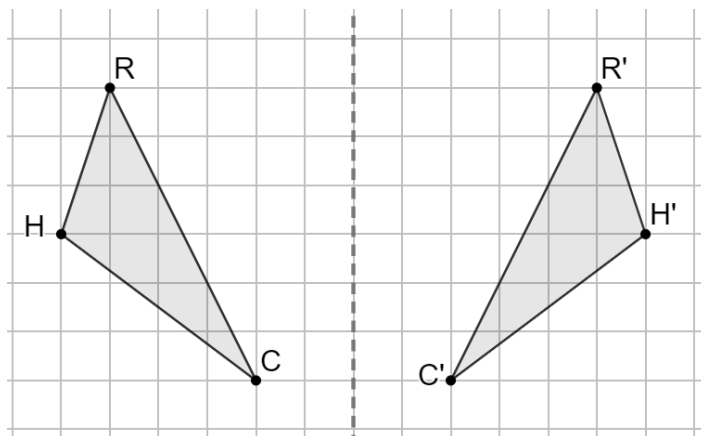


	Transformation	Triangle
8.	$(x, y) \rightarrow (x - 7, y + 5)$	$N'H'R'$
9.	$(x, y) \rightarrow (-y, x)$	$N''H''R''$
10.	$(x, y) \rightarrow (x + 11, y + 4)$	$N'''H'''R'''$

Geometry
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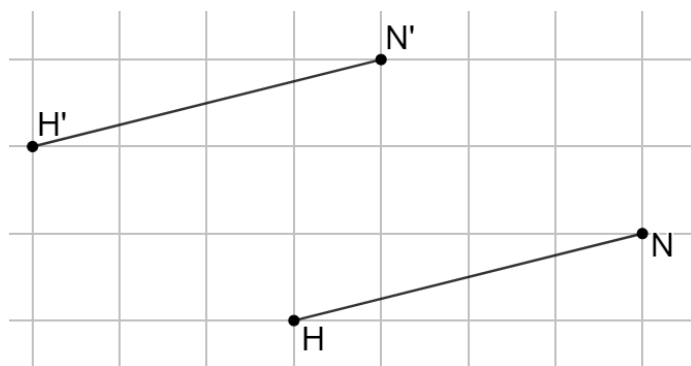
Triangles RHC and $R'H'C'$ are shown on grid paper.



1. Explain what method(s) can be used to determine if each angle of $\triangle RHC$ has the same measure as $\triangle R'H'C'$.
2. Explain what method(s) can be used to determine if each side of $\triangle RHC$ has the same length as $\triangle R'H'C'$.
3. Explain why the transformation of $\triangle RHC$ shown on the grid can be classified as isometry.

Line segments HN and $H'N'$ are shown on the grid.

4. Show the mapping of point H to H' .
5. Describe the transformation applied to $\overline{HN}$ that resulted in $\overline{H'N'}$.
6. Complete the statement.



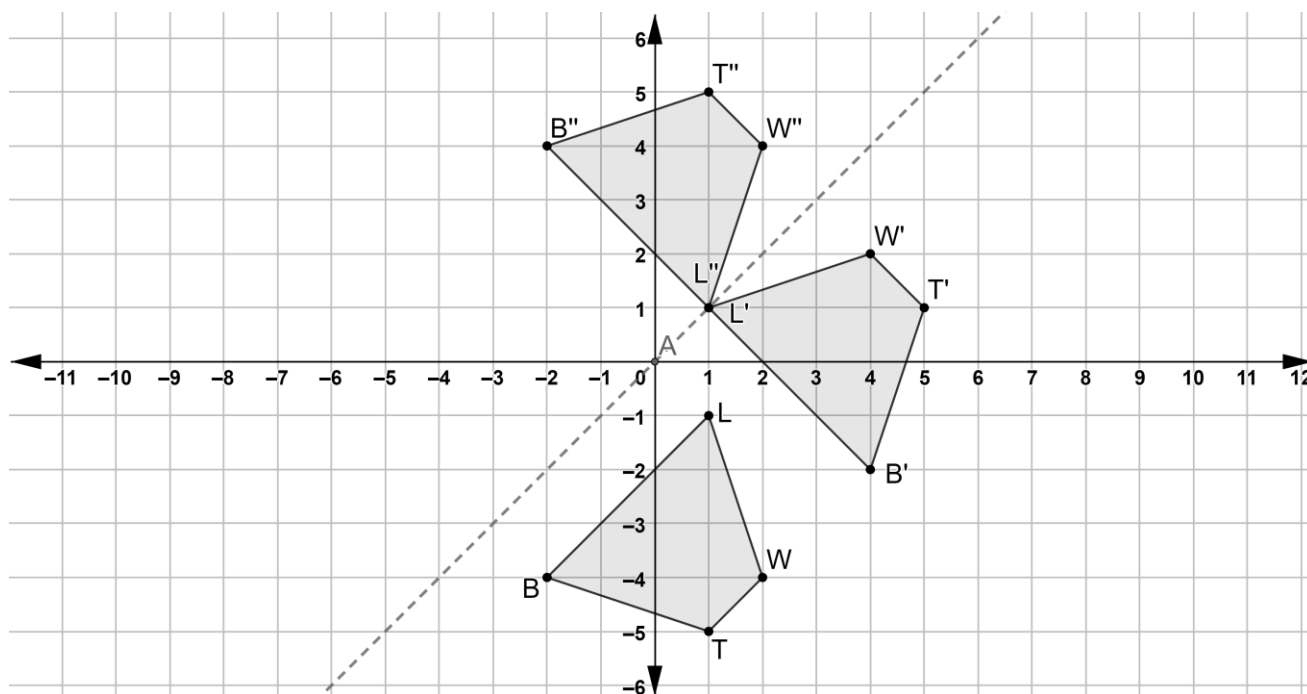
- Each point on
- Only the endpoints of

$\overline{HN}$ can be mapped to

- a corresponding point on
- the endpoints of

$\overline{H'N'}$.

Quadrilaterals $LWTB$, $L'W'T'B'$, and $L''W''T''B''$ are shown on the coordinate grid.



7. Describe what movements should be done with a patty paper copy of quadrilateral $LWTB$ so that the copy of $LWTB$ lays on top of $L'W'T'B'$.
8. Describe what movements should be done with a patty paper copy of quadrilateral $L'W'T'B'$ so that the copy of $L'W'T'B'$ lays on top of $L''W''T''B''$.
9. Complete the statements.

The transformations used in both sequences

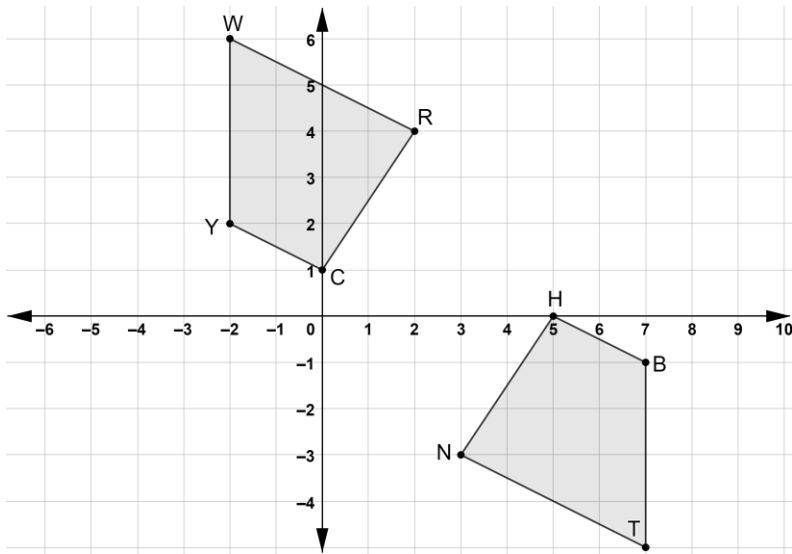
☐ are
☐ are not

examples of isometry. All of the points of

☐ $L'W'T'B$
☐ $L''W''T''B''$
☐ $L'W'T'B'$ and $L''W''T''B''$

can be mapped to all of the points of $LWTB$.
10. Use the definition of congruence in terms of rigid motions to explain how both sequences of transformations for quadrilateral $LWTB$ resulted in $LWTB \cong L''W''T''B''$.

Quadrilaterals $CYWR$ and $BTNH$ are shown on the coordinate grid.

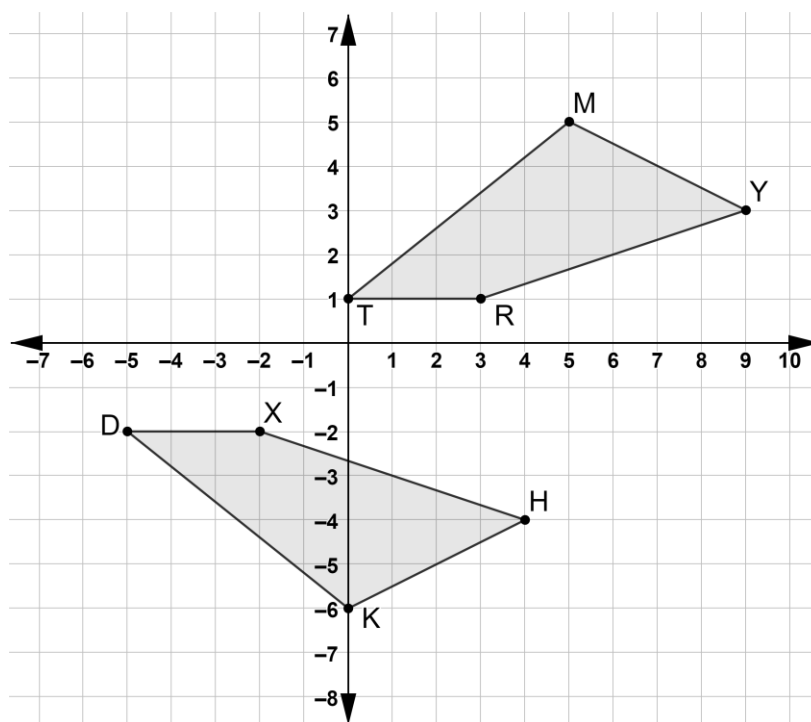


- Describe a sequence of transformations that will map quadrilateral $CYWR$ onto quadrilateral $BTNH$.
- The sequence of transformations that will map quadrilateral $CYWR$ onto quadrilateral $BTNH$ includes a rotation. Describe the sequence using a different rotation.
- Complete the table that shows which vertices were mapped to each other when quadrilateral $CYWR$ was mapped to quadrilateral $BTNH$.

Quadrilateral $CYWR$	Quadrilateral $BTNH$
C	
Y	
W	
R	

- Write the algebraic description that matches the written description one of the transformations.
 $(x, y) \rightarrow$ _____
 $(x, y) \rightarrow$ _____
- Complete the congruence statement. $CYWR \cong$ _____

Quadrilaterals $KHXD$ and $TMYR$ are shown on the coordinate grid.



6. Describe a sequence of transformations that will map quadrilateral $KHXD$ onto quadrilateral $TMYR$.

7. Show the mapping of the transformations on the coordinate grid.

8. Complete the statement.

Vertex D maps to vertex _____, vertex K maps to vertex _____, vertex H maps to vertex _____, and vertex X maps to vertex _____.

9. Write the algebraic description that matches the written description of the transformations.

$$(x, y) \rightarrow \underline{\hspace{2cm}}$$

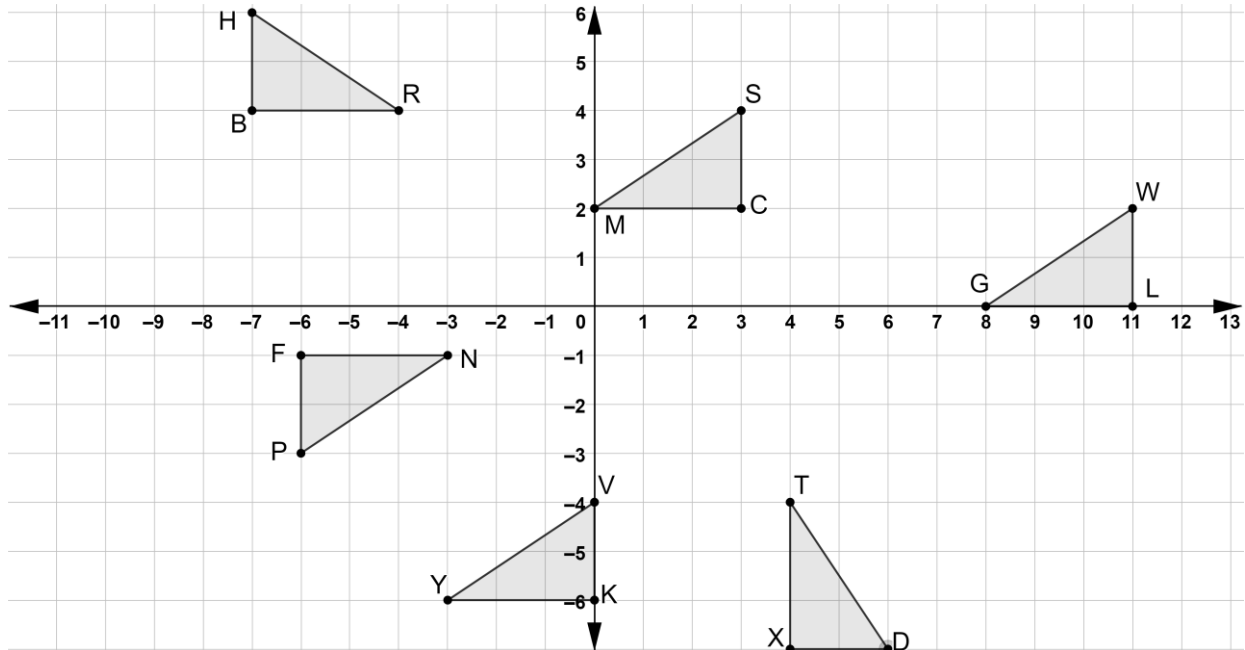
$$(x, y) \rightarrow \underline{\hspace{2cm}}$$

10. Complete the congruence statement. $KHXD \cong \underline{\hspace{2cm}}$

Geometry
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Six triangles are shown on the coordinate grid.



Give a written description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

1. $\triangle FNP \cong \triangle CMS$

2. $\triangle FNP \cong \triangle XTD$

3. $\triangle FNP \cong \triangle KYV$

4. $\triangle FNP \cong \triangle BRH$

5. $\triangle FNP \cong \triangle LGW$

Give an algebraic description of the transformations that lead to the given congruence statement where $\triangle FNP$ is the preimage.

6. $\triangle FNP \cong \triangle CMS$

7. $\triangle FNP \cong \triangle XTD$

8. $\triangle FNP \cong \triangle KYV$

9. $\triangle FNP \cong \triangle BRH$

10. $\triangle FNP \cong \triangle LGW$